Practical 2.1: CRD Installation with Docker

Clone, configure, build and launch a complete CRD (GeoNode) instance

Day 2 · 3 hours · Resilience Academy SDI Training

Reference: `crd.resilienceacademy.ac.tz` | Data: Tanga & Mwanza, Tanzania

Learning Objectives:

- Clone and configure the GeoNode project template
- Build and launch a complete CRD stack using Docker Compose
- Access and administer your local CRD instance
- Compare your local installation to the live crd.resilienceacademy.ac.tz

Prerequisites

- Docker and Docker Compose installed (Practical 1.1 complete)
- At least 6GB RAM available to Docker
- 10GB free disk space
- Internet connection for pulling images

Docker memory: Open Docker Desktop settings (if on Windows/Mac) and increase memory to at least 6GB before starting.

Step 1: Clone GeoNode Project

```
# Clone the official GeoNode project template
git clone https://github.com/GeoNode/geonode-project.git my-crd
cd my-crd

# Check out a stable release tag
git checkout tags/4.2.0

# View project structure
ls -la
cat README.rst | head -30
```

Step 2: Create Python Environment and Scaffold

```
# Create virtual environment
python3 -m venv .venv
source .venv/bin/activate

# Install Django for scaffolding
pip install "Django>=3.2,<4.0"

# Create project scaffold from template
django-admin startproject \
  --template=. \
```

```
  --extension py,rst,json,yml,ini,env \
  my_geonode .

# Verify structure created
ls my_geonode/
```

Step 3: Configure Environment File

```
# Copy sample .env file
cp .env.sample .env

# Edit key settings
nano .env
```

Edit these values in `.env`:

```
# Site URL (must end with /)
SITEURL=http://localhost/

# Database password (change this!)
POSTGRES_PASSWORD=resilience2024

# GeoServer admin password
GEOSERVER_ADMIN_PASSWORD=geoserver2024

# Secret key (generate a random string)
SECRET_KEY=my-very-secret-key-for-training-2024

# Email (optional for training)
DJANGO_EMAIL_HOST=localhost
```

Step 4: Build and Start

```
# Build all Docker images (takes 5-10 minutes first time)
docker compose build

# Start all services in background
docker compose up -d

# Watch the startup logs (takes 3-5 minutes)
docker compose logs -f
# Wait until you see:
# django_1 | Starting development server at http://0.0.0.0:8000/

# Check all containers are running
docker compose ps
```

Step 5: Initialize CRD

```
# Run database migrations
docker compose exec django python manage.py migrate

# Create admin superuser
docker compose exec django python manage.py createsuperuser
# Enter: username=admin, email=admin@example.com, password=Admin1234!
```

```
# Load default fixtures
docker compose exec django python manage.py loaddata \
  initial_data.json 2>/dev/null || true

# Sync GeoServer layers
docker compose exec django python manage.py sync_geonode_maps
```

Step 6: Access Your CRD

```
# Test that CRD is responding
curl -I http://localhost
# Should return: HTTP/1.1 200 OK

# Open in browser:
# http://localhost - Main CRD interface
# http://localhost/en/admin - Django admin
# http://localhost/geoserver - GeoServer admin
```

Compare your local CRD with the live platform: https://crd.resilienceacademy.ac.tz

Troubleshooting

```
# Port 80 already in use?
sudo lsof -i :80
# Change NGINX_PORT in .env to 8000, then docker compose restart

# GeoServer not connecting?
docker compose logs geoserver | tail -20
docker compose restart geoserver

# Out of memory?
docker stats # see container memory usage
# Increase Docker memory limit to 6GB+
```

Checkpoint:

- `docker compose ps` shows all services as "running" or "Up"
- `http://localhost` opens the CRD web interface
- You can login with admin account
- GeoServer is accessible at `http://localhost/geoserver`
- `http://localhost/api/v2/datasets/` returns JSON